Cheap-tier models and hosting: comparison (2026-10-22)

The question: which model and hosting should serve the router's simple tasks? The options are a cheap closed model now, managed open-weight, or self-hosted on GCP when traffic grows. This doc also covers when each choice makes sense. Prices were checked on 2026-10-22; sources are listed at the end.

1. What one market run uses (measured, run `mrun_20261022T083528`, Kozmo, through buyers)

Task (simple → cheap tier)	Calls	Avg input tokens	Avg output tokens	gpt-4o-mini stand-in result
vichara.v1 (answer from the whitepaper, with exact quotes)	19	9,800	130	5 kept, 14 escalated (10 abstention, 4 grounding)
listing_extraction.v2 (companies on a web page, quoted)	45	6,000	260	25 kept, 20 escalated (excerpts, link IDs)
link_verification.v1	4	12,900	770	4 kept
buyer_roles.v2, query_plan.v2, homepage_identity.v2	7	600–2,900	30–200	7 kept

- **Cheap tier per run:** about **515K input** / **20K output tokens**. The load is almost all input (reading pages and the PDF), with very little writing.
- **Whole run:** $1.83, against $2.55 if every call went to frontier. **Frontier is $1.74 of the $1.83.** Choosing a cheap model therefore saves cents per run. Cutting escalations, or choosing the next frontier model, moves dollars.

2. Deadline that changes the plan

OpenAI removes **gpt-4o, gpt-4o-mini, gpt-4.1 and gpt-4.1-mini from its API on Jan 14, 2027** (announced Oct 14). The named replacement for the small models is GPT-6 Luna.

- "Start with 4.1 mini" would last about 12 weeks and costs 4× Luna.
- Our frontier model (gpt-4o) must move by the same date. GPT-6.1 Sol is the recommended replacement and works with the current client.

3. Options compared

Cost per run covers the cheap tier only, using the token profile above. Escalations to frontier are extra and the same for every option.

Option	Cost per run	Pros	Cons
A. Cheap closed: GPT-6 Luna (or Haiku 5.5)	**$0.06**	No new account (OpenAI key we have). No ops. Cheapest today	Not open-weight: no portability, vendor can retire it like 4.1 mini. Quality on our checks not yet tested

A′. GPT-4.1 mini	$0.24	Known model	Retires Jan 14, 2027; 4× Luna's price
B. Managed open-weight (Fireworks / Together / DeepInfra)	**$0.08** (GLM-5.3-Flash for quoting and extraction, gpt-oss-120b for light tasks)	Pay per token, nothing when idle. Model can move between hosts or in-house later. Fireworks keeps no prompts by default	New account and key. Hosts drop models (Vertex removed 16 on Oct 21). Reasoning must be switched off on GLM
B′. Managed on GCP (Vertex AI pay-per-token)	$0.05 (gpt-oss-120b for everything) / $0.16 (DeepSeek V4.1 Flash + gpt-oss)	GCP billing; same project as Firebase	Small choice: only DeepSeek V4.1 Flash, gpt-oss-120b and Gemma 4 26B remain. Strict JSON output per model not verified
C. Self-hosted on GCP, always on	Fixed monthly cost (table 4)	Full control, data stays in our project, fixed cost at high volume	Pays per hour with no traffic. GPU quota approval needed. Setup takes 1–3 weeks, plus 2–8 h/month upkeep
C′. Self-hosted, scale to zero (Cloud Run GPU)	$0.21–1.12	Nothing paid when idle	2–5 min cold start; about 15 min idle time billed after each burst. Costs 2–12× managed per run

Which models fit one GPU. Only gpt-oss-120b (one 80–96 GB GPU) and Gemma 4 31B (one L4 at 4-bit) can be self-hosted on a single GPU. GLM-5.3-Flash, MiMo-V2.6-Flash and DeepSeek V4.1 Flash are 300–550B MoE models. They need an 8-GPU node at about $63K/month, which rules them out for self-hosting.

4. Monthly cost at traffic levels (cheap tier only)

Self-host figures exclude upkeep (about $200–800/month of engineering time).

Runs/ day	A. Luna	A′. 4.1 mini	B. Managed open-weight	B′. Vertex gpt-oss	C. RTX PRO 6000 Spot + gpt-oss (about 800 runs/ day per GPU)	C. L4 Spot + Gemma 4-bit (about 130 runs/ day per GPU)
5 (today)	$9	$35	$12	$8	$1,190	$309
50	$92	$354	$125	$81	$1,190	$309
500	$915	$3,540	$1,245	$810	$1,190	$1,236 (4 GPUs)
2,000	$3,660	$14,160	$4,980	$3,240	$3,570 (3 GPUs)	$4,635 (15 GPUs)

- **Break-even for self-hosting.** It beats managed open-weight only

above about **400–600 runs/day**, held steady, at over 50% GPU use. Even then it only roughly matches Luna or Vertex gpt-oss-120b once upkeep is counted.
- **GPU types that never win.** On-demand L4, A100 and H100 cost more per run than managed APIs even at 100% use.

5. Recommendation

1. **Now (a few runs/day):** pay per token; do not self-host.
    - **Trial:** run the recorded Kozmo simulation with GLM-5.3-Flash (Fireworks) and with GPT-6 Luna.
    - **Decision rule:** choose the model with the fewest escalations, not the lowest token price. Each escalation adds about 15–30× the cheap call's cost.
2. **Per task, not one model:**

Task group	Primary	Challenger
Quoting from long documents (vichara)	GLM-5.3-Flash	DeepSeek V4.1 Flash
Page extraction (listing, homepage)	GLM-5.3-Flash	MiMo-V2.6-Flash
Light JSON tasks (links, buyers, query plan)	gpt-oss-120b	Gemma 4 31B

    The router already keeps one open-weight endpoint. Choosing the model per task would be a small change in `config/routing.yaml` and needs sign-off.
3. **Keep two providers** (for example Fireworks plus Vertex or DeepInfra), because hosts retire models with little notice.
4. **Look at self-hosting again** only if traffic holds above about 400 runs/day or if data residency requires it. The candidate then is an RTX PRO 6000 (g4) running gpt-oss-120b, on Spot or a 1-year

commitment.

5. **Before Jan 14, 2027:** move frontier from gpt-4o to GPT-6.1 Sol, and drop gpt-4o-mini as the simulation stand-in.

6. Not verified

- **Vertex:** the official pricing page did not load; gpt-oss-120b is quoted at both $0.09 and $0.15 per 1M input tokens. Strict JSON output per Vertex model is not confirmed.
- **Fireworks:** one serving path lists GLM-5.3-Flash at $2.96/$7.41; the model page says $0.15/$0.50.
- **Self-host throughput:** figures are estimates (±50%), because public benchmarks use short prompts.
- **Cloud Run:** the real length of the idle time billed after each burst is not measured.
- **Not tested yet:** none of the open-weight models or Luna has been run on our tasks.

Sources (seen 2026-10-22)

- OpenAI deprecations: https://developers.openai.com/api/docs/deprecations
- OpenAI pricing: https://developers.openai.com/api/docs/pricing
- Anthropic pricing: https://platform.claude.com/docs/en/about-claude/pricing
- Fireworks pricing: https://docs.fireworks.ai/serverless/pricing
- Fireworks GLM-5.3-Flash: https://fireworks.ai/models/fireworks/glm-5p3-flash
- Fireworks data handling: https://docs.fireworks.ai/guides/security_compliance/data_handling
- Together GLM-5.3-Flash: https://www.together.ai/models/glm-5-3-flash
- DeepInfra pricing: https://deepinfra.com/pricing
- OpenRouter model endpoints: https://openrouter.ai/api/v1/models/

z-ai/glm-5.3-flash/endpoints

- Vertex open models: https://docs.cloud.google.com/vertex-ai/generative-ai/docs/maas/use-open-models
- Vertex retirements: https://www.developersdigest.tech/blog/vertex-ai-shutdowns-october-2026
- Vertex prices: https://calcaas.com/llm-pricing/vertex-ai
- GCP VM prices: https://gcloud-compute.com/g4-standard-48.html, https://gcloud-compute.com/g2-standard-4.html, https://gcloud-compute.com/a3-highgpu-1g.html
- Cloud Run GPU: https://docs.cloud.google.com/run/docs/configuring/services/gpu
- Cloud Run cold starts: https://cloud.google.com/blog/topics/developers-practitioners/a-guide-to-ai-cold-starts-on-cloud-run
- Model sizes: https://recipes.vllm.ai/zai-org/GLM-5.3-Flash, https://recipes.vllm.ai/deepseek-ai/DeepSeek-V4.1-Flash
- gpt-oss-120b on H100: https://inferencex.semianalysis.com/run/gptoss-120b-on-h100